

AYUSH PATEL

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EDUCATION

McMaster University

Hamilton, ON

Bachelor of Engineering in Software Engineering (CO-OP) - GPA: 3.9/4

Expected Graduation: May 2027

- Coursework: Software Design I & II | Software Architecture | Object-Oriented Programming | Data Structures & Algorithms | Databases | Software Testing | Requirements Engineering & Security Considerations | Computer Architecture | Digital Systems | Discrete Mathematics I & II | Statistics

Awards: NSERC USRA | McMaster Award of Excellence | Frank Sherman Scholarship

EXPERIENCE

Mobile Application Security Research Intern

May 2025 - Present

McMaster Center for Software Certification (McSCert)

Hamilton, ON

- Investigating the security vulnerabilities of mobile healthcare applications

Software Developer

Dec 2024 - Present

iBiomed Society - McMaster

Hamilton, ON

- Developing a cross-platform mobile app for patients with scleroderma using **React Native** and **Firebase**, providing financial and mental health resources for **1000+ users**
- Developed a dynamic, responsive website reaching **500+ users** utilizing **Next.js** and **Tailwind CSS**, integrating external APIs to deliver real-time content and interactive user experiences
- Facilitated effective team communication and task management using **Jira** to create, track, and **manage user stories** and bug fixes within a collaborative **Agile environment**.
- Developed **comprehensive technical documentation** including system design considerations and API specifications, reducing onboarding time and ensuring consistent development standards

Embedded Systems Developer

Feb 2025 - Present

Computer Vision Sub Team - McMaster RoboSub

Hamilton, ON

- Developing real-time object detection systems using **PyTorch** and **OpenCV** frameworks, optimizing algorithms for autonomous underwater vehicle operation in diverse aquatic conditions
- Building embedded vision solutions utilizing **NVIDIA Jetson hardware**, implementing ROS-based pipelines to integrate with AUV control systems for obstacle avoidance and competition task execution
- Creating **comprehensive technical documentation** and system architecture specifications to facilitate knowledge transfer across interdisciplinary sub-teams and ensure consistent development standards

PROJECTS

MovieWhiz | *React Native, Express.js, RESTful API, Jest, Android Studio, Expo, UML, Firebase*

Jan 2025 - May 2025

- Led the design and development of MovieWhiz, an **Android mobile application** that helps users identify forgotten movies using multiple expert systems including LLM's, soundtrack recognition and comprehensive movie databases.
- Managed the **full software development lifecycle**, including requirements gathering, system architecture design, UML diagrams, implementation, and testing, ensuring project coherence and quality.
- Developed a **RESTful API** backend using **Express.js** and tested it using **Jest** and Supertest, ensuring robust and reliable application functionality as well as data management using **Firestore**
- Designed for **scalability and extensibility**, enabling seamless integration of additional expert systems, LLM's and databases.

RescueMission | *Java, Apache Maven, JUnit, UML, GitHub*

Jan - Mar 2024

- Designed and implemented a **rescue drone control program** to locate stranded individuals and identify optimal rescue points, integrating efficient battery management algorithms to maximize operation time.
- Applied **SOLID principles** and **GoF design patterns** to build a scalable, maintainable system, leveraging object-oriented design and robust unit testing with **JUnit** to ensure functionality and reliability
- Utilized a **Kanban board** within an **Agile environment**, delivering an MVP within 2 weeks and iteratively improving based on feedback

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, Bash, Verilog, ARM64 Assembly

Developer Tools and Frameworks: Git, GitHub, Linux, React, Next.js, Express.js, Node, Tailwind CSS, Apache Maven, JUnit, Jupyter Notebook, Visual Studio Code, IntelliJ, React Native, Firebase, Jest, UML, Android Studio, Expo, ROS2, Jira, Kanban

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, CloudWatch, API Gateway), Docker, Kubernetes, AWS CLI, GitHub Actions

Libraries: Pandas, NumPy, TensorFlow, PyTorch, OpenCV, Scikit-learn, Matplotlib

Professional Certifications: AWS Certified Cloud Practitioner (May 2025). Introduction to Containers w/ Docker, Kubernetes & OpenShift (May 2025)